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FINALS ACTIVITY 2 - Quality Assurance (QA) vs Quality Control (QC)

1. What is Quality Assurance (QA)?

Quality Assurance (QA) is a proactive process focused on preventing defects by ensuring that the processes used to develop a product or service are effective and consistent. It is a systematic approach to building quality into the product from the beginning, emphasizing planning, process improvement, and adherence to standards. QA aims to ensure that the development or production process itself is capable of delivering a high-quality outcome.

- **Focus:** Process-oriented.
- **Goal:** Prevent defects before they occur.
- **Example:** Defining coding standards, conducting process audits, or implementing training programs.

2. Characteristics of Quality Assurance

- **Preventive:** Aims to avoid issues by improving processes rather than fixing defects after they occur.
- **Process-Driven:** Focuses on the methods, tools, and techniques used to create the product.
- **Continuous Improvement:** Involves regular evaluation and refinement of processes.
- **Proactive Approach:** Identifies potential risks and addresses them early in the development lifecycle.
- **Team Responsibility:** Involves everyone in the organization, not just a specific team.
- **Standards-Based:** Relies on predefined quality standards (e.g., ISO 9001) and best practices.
- **Long-Term:** Focuses on sustained quality over time rather than immediate results.

3. What is Quality Control (QC)?

Quality Control (QC) is a reactive process that involves inspecting, testing, and verifying the finished product or service to ensure it meets specified requirements and quality standards. It focuses on identifying and correcting defects in the final output before it reaches the end user.

- **Focus:** Product-oriented.
- **Goal:** Detect and fix defects in the deliverables.

- **Example:** Running test cases on a software application, inspecting manufactured goods for flaws.

4. Characteristics of Quality Control

- **Reactive:** Identifies and corrects defects after they occur in the product.
- **Product-Focused:** Centers on the output rather than the process used to create it.
- **Inspection-Based:** Relies on testing, reviews, and checks to verify quality.
- **Specific Team Responsibility:** Typically performed by a dedicated team (e.g., testers or inspectors).
- **Short-Term:** Aims to ensure the immediate quality of a specific product or batch.
- **Standards Compliance:** Ensures the product meets predefined quality criteria or client requirements.
- **Corrective Actions:** Involves fixing issues found during inspection or testing.

5. Difference Between Quality Assurance (QA) and Quality Control (QC)

Aspect	Quality Assurance (QA)	Quality Control (QC)
Definition	A process to ensure quality in how a product is made.	A process to verify the quality of the final product.
Focus	Process-oriented (how the work is done).	Product-oriented (what is produced).
Approach	Proactive (prevents defects).	Reactive (detects and corrects defects).
Goal	Build quality into the process.	Ensure the product meets quality standards.
Timing	Throughout the development lifecycle.	After development or during final stages.
Responsibility	Involves the entire team or organization.	Typically handled by a specific team (e.g., testers).
Activities	Process audits, training, defining standards.	Testing, inspection, defect reporting.

Example	Creating a testing methodology for software.	Executing test cases to find bugs in software.
Outcome	Consistent, reliable processes.	Defect-free product delivered to the user.
Scope	Broad, systemic improvements.	Narrow, specific to the product being evaluated.